

Using Encryption to Enhance Confidentiality and Integrity (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 05

Student:
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Time on Task:
11 hours, 31 minutes

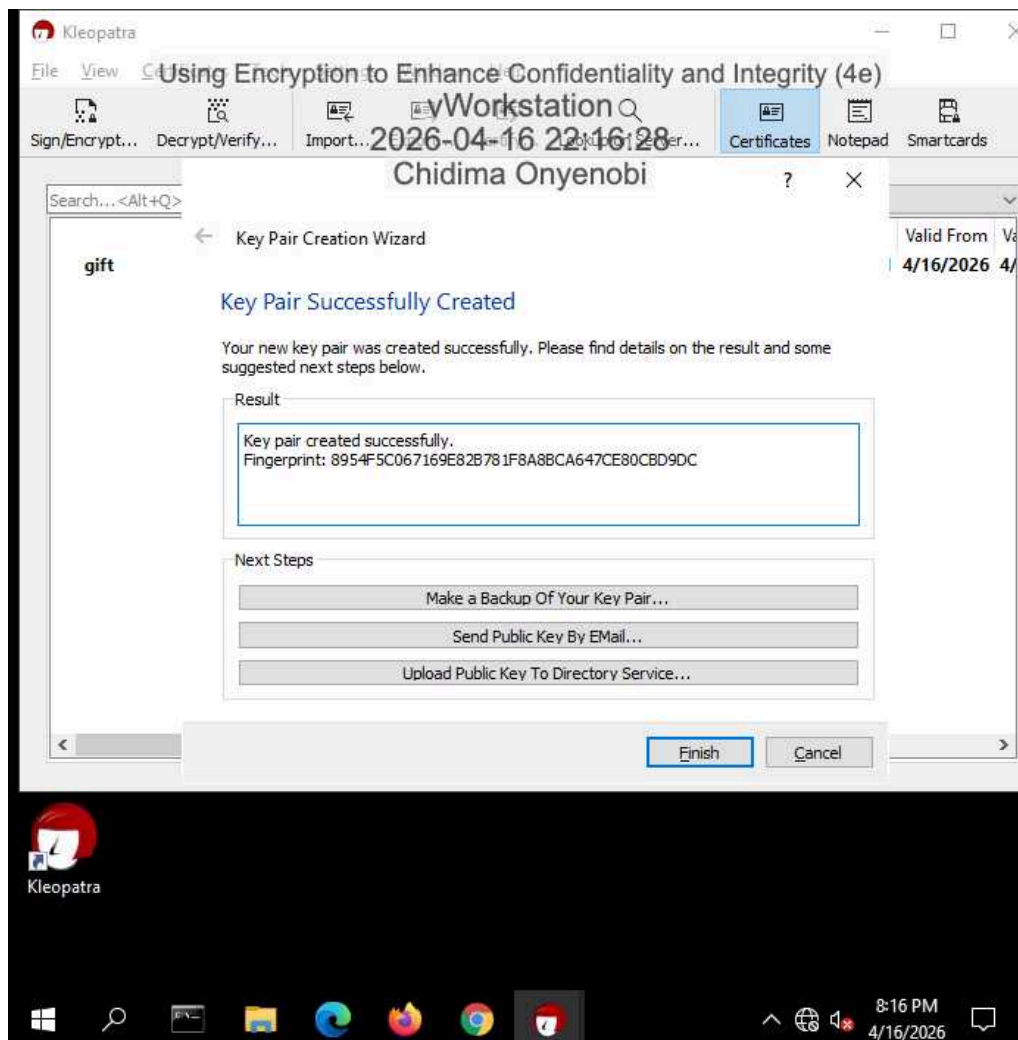
Progress:
100%

Report Generated: Friday, April 17, 2026 at 1:40 AM

Section 1: Hands-On Demonstration

Part 1: Create and Exchange Asymmetric Encryption Keys

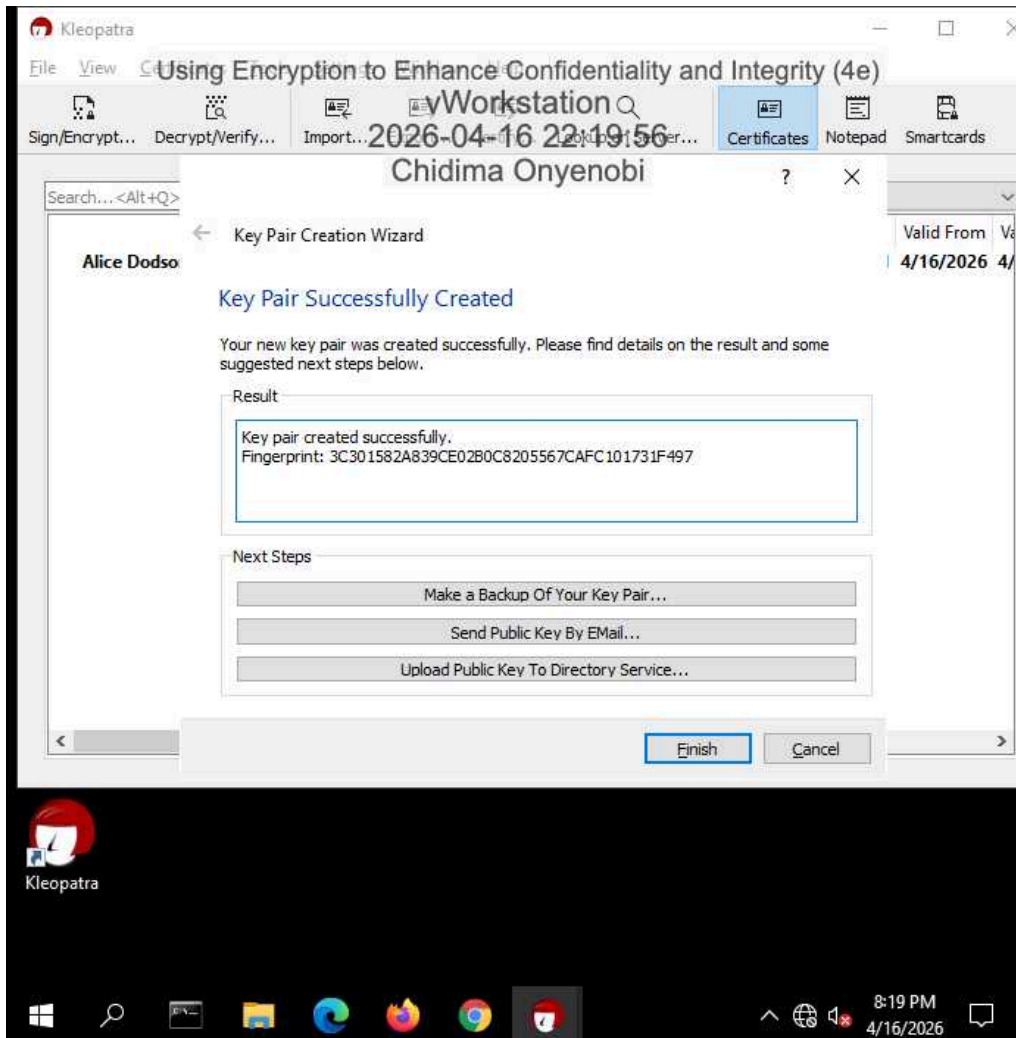
9. Make a screen capture showing the **fingerprint** for your key pair.



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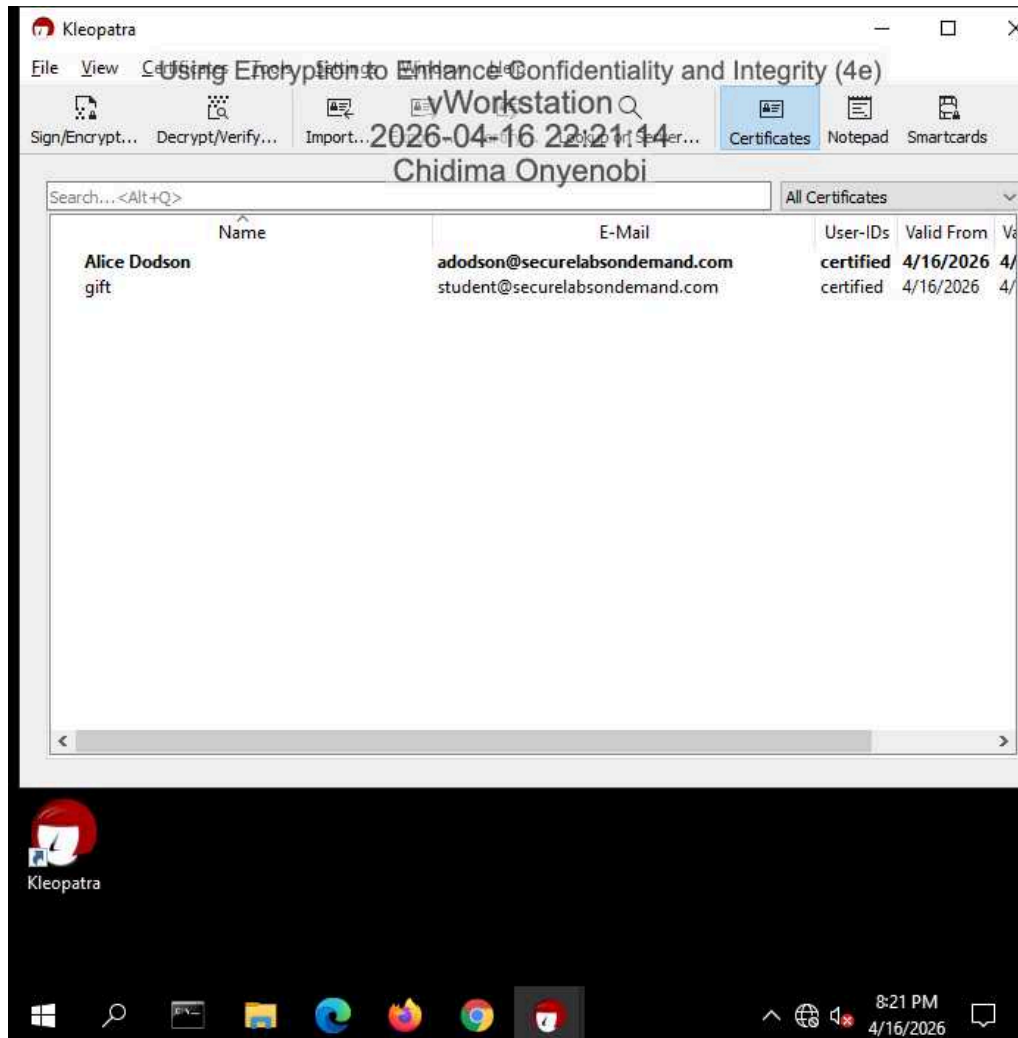
22. Make a screen capture showing the **fingerprint** for Alice's key pair.



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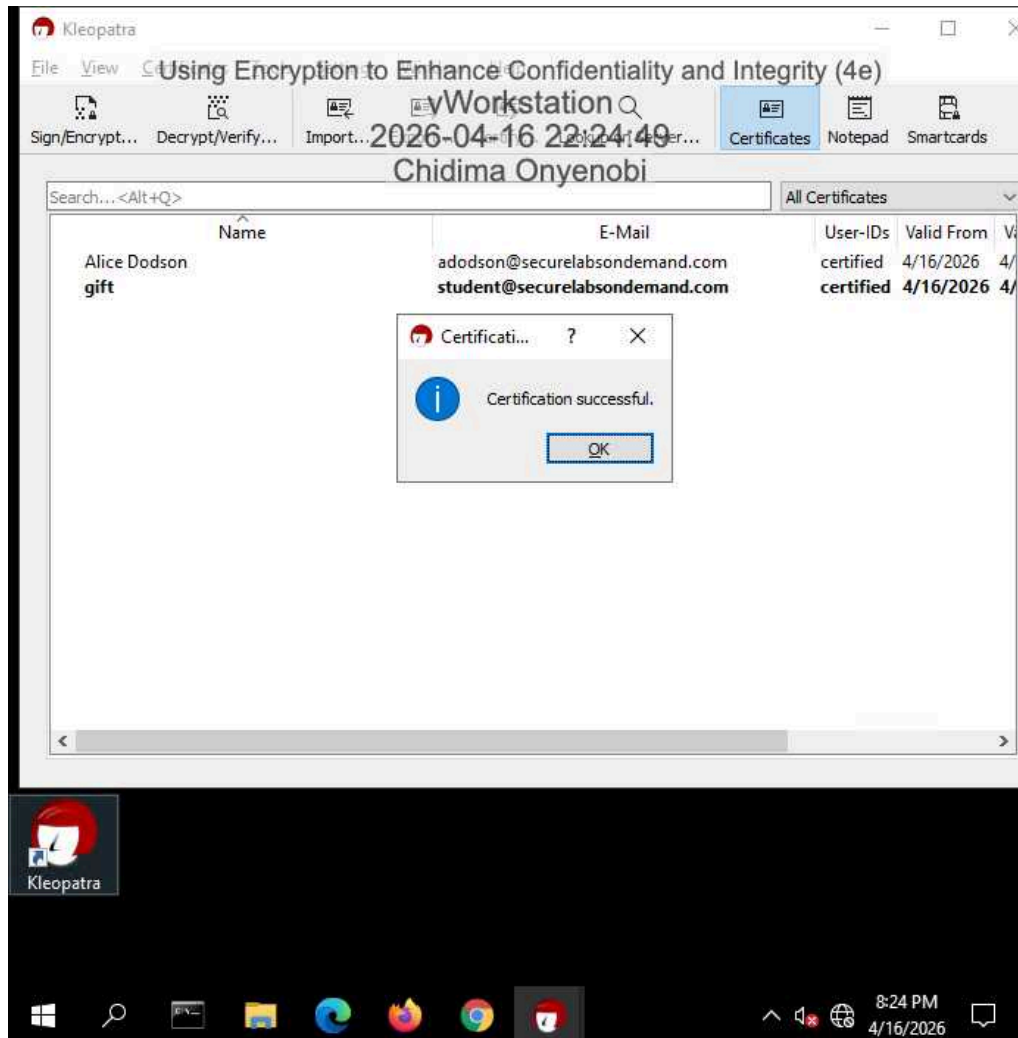
30. Make a screen capture showing your public key in Alice's certificate cache.



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35. Make a screen capture showing Alice's public key in your certificate cache.

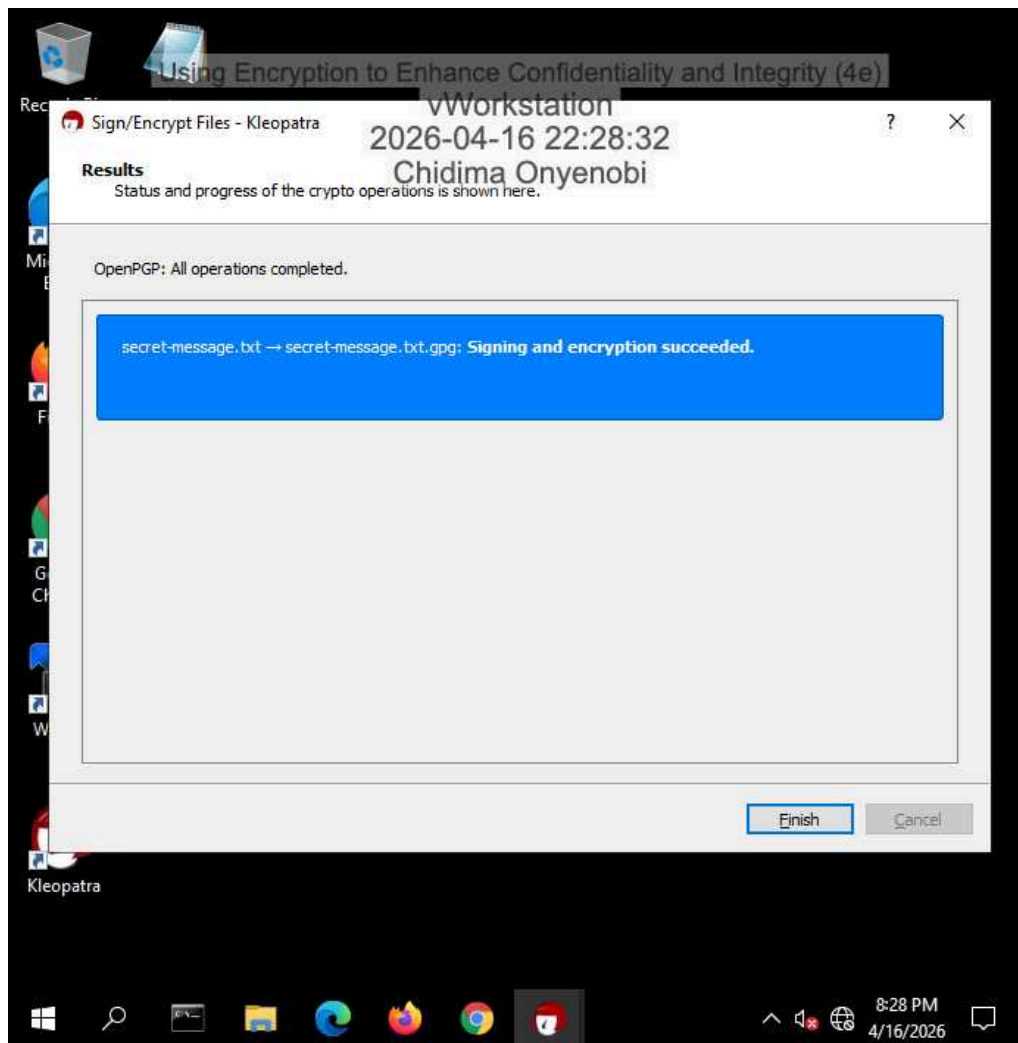


Part 2: Encrypt a File Using Asymmetric Encryption

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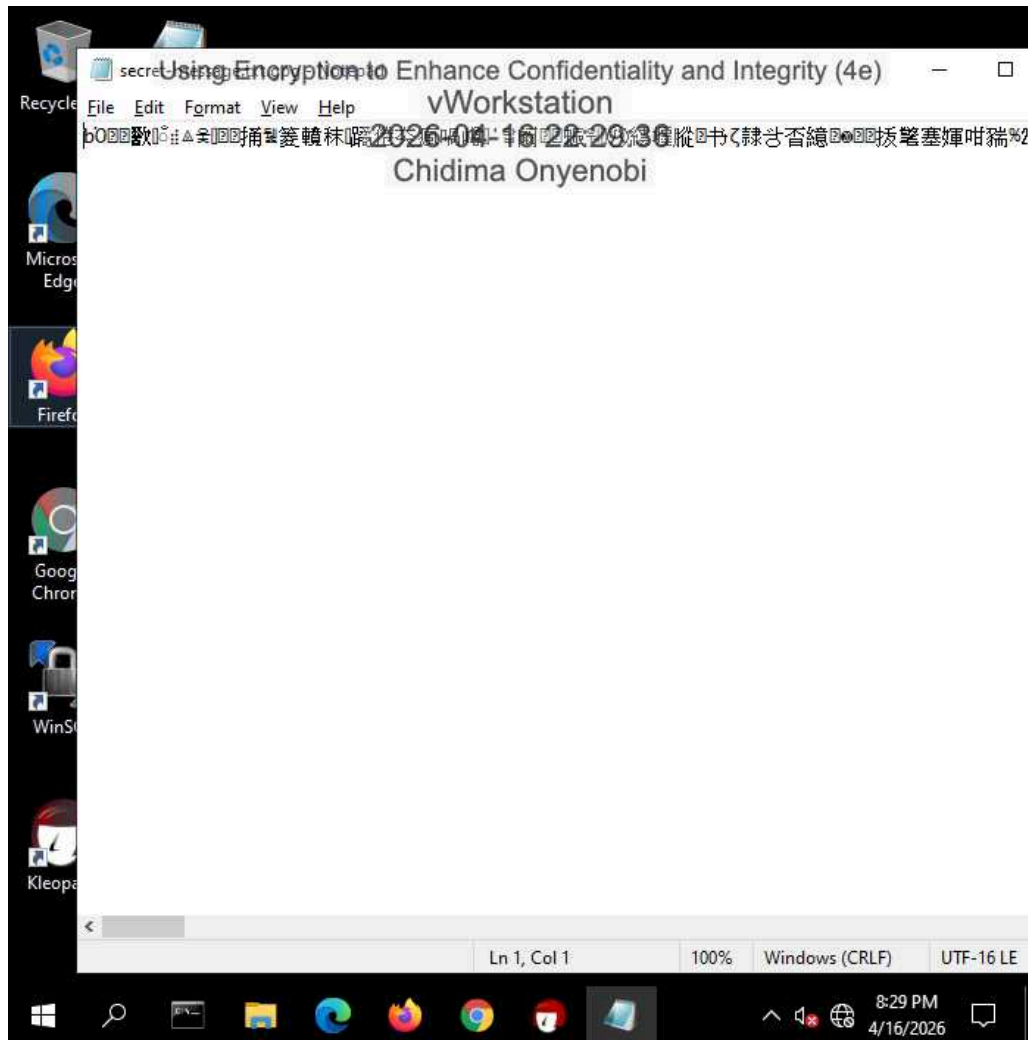
9. Make a screen capture showing the **successful signing and encryption message**.



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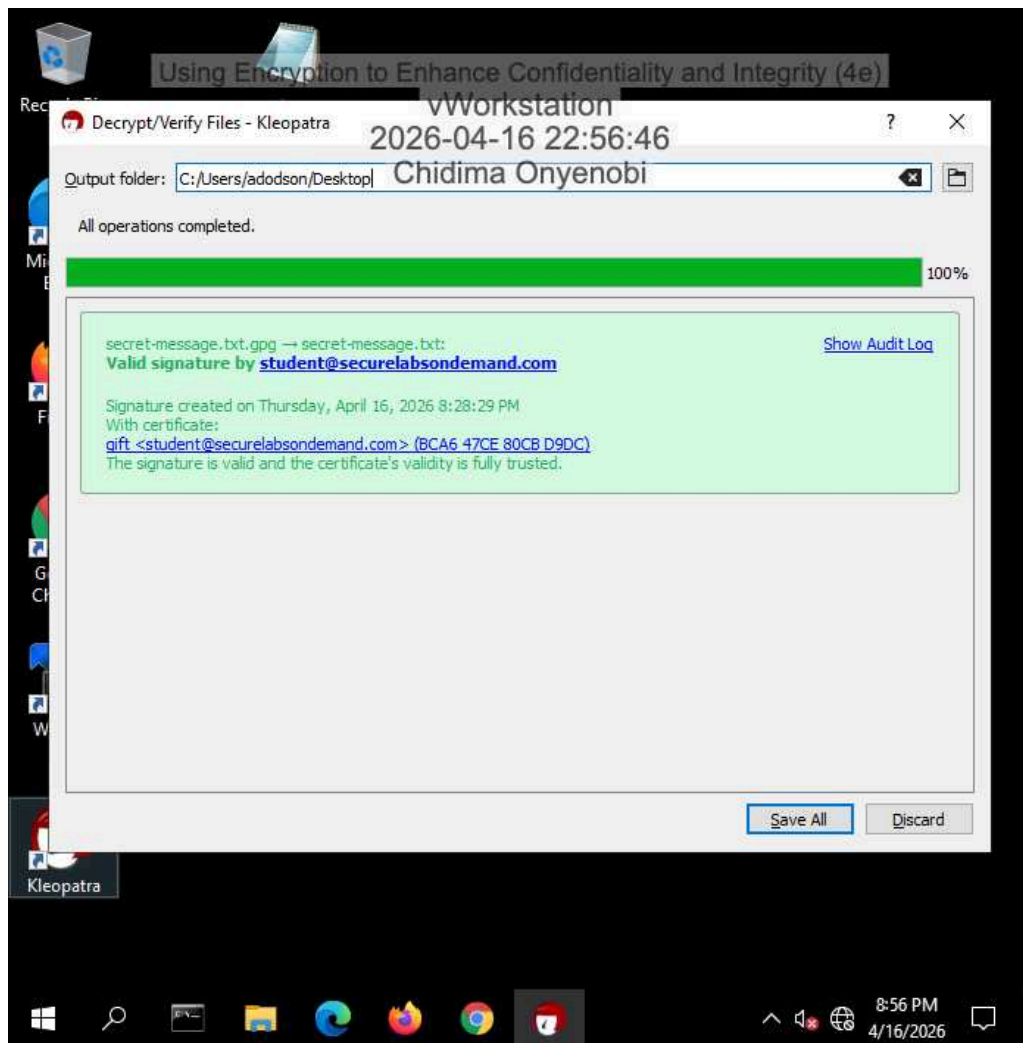
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12. Make a screen capture showing the **ciphertext**.



Part 3: Decrypt a File Using Asymmetric Encryption

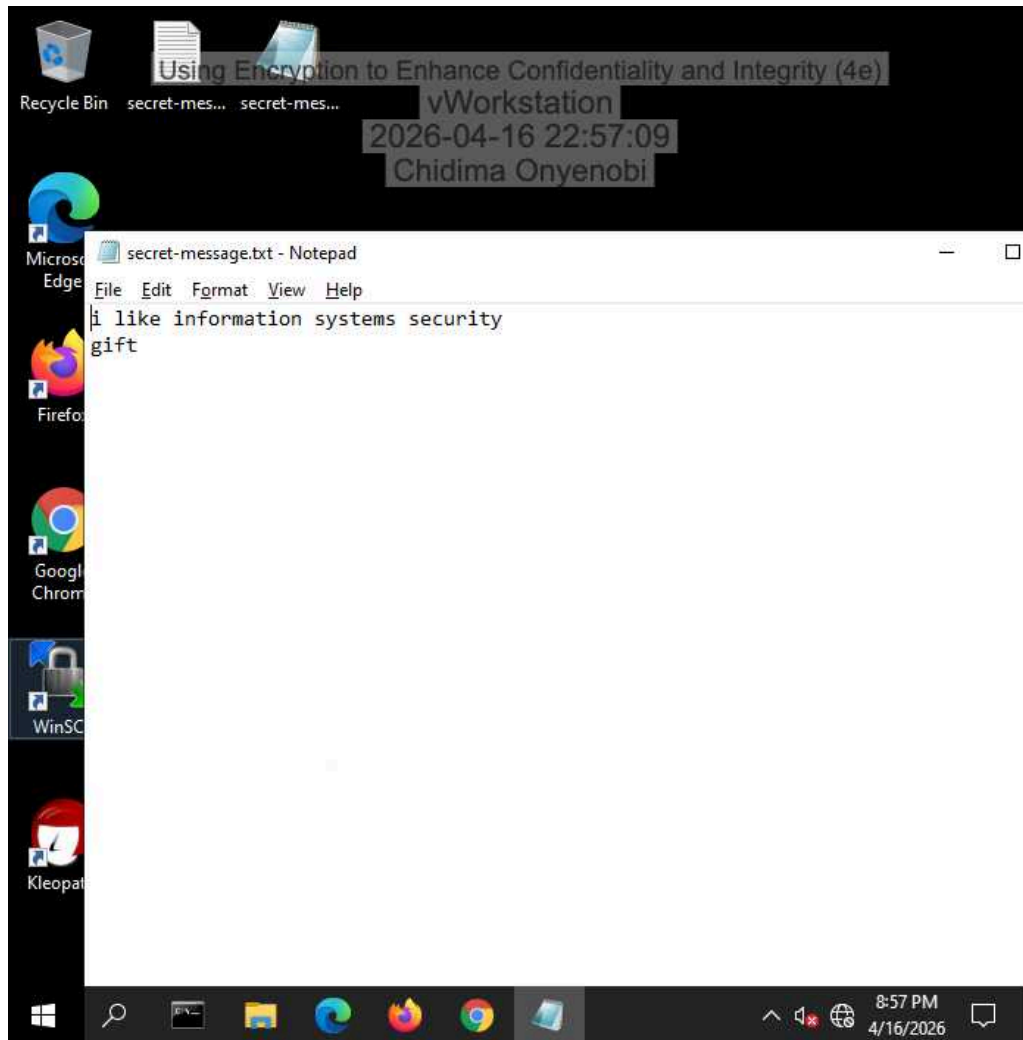
15. Make a screen capture showing the **Decrypt/Verify Files** window.



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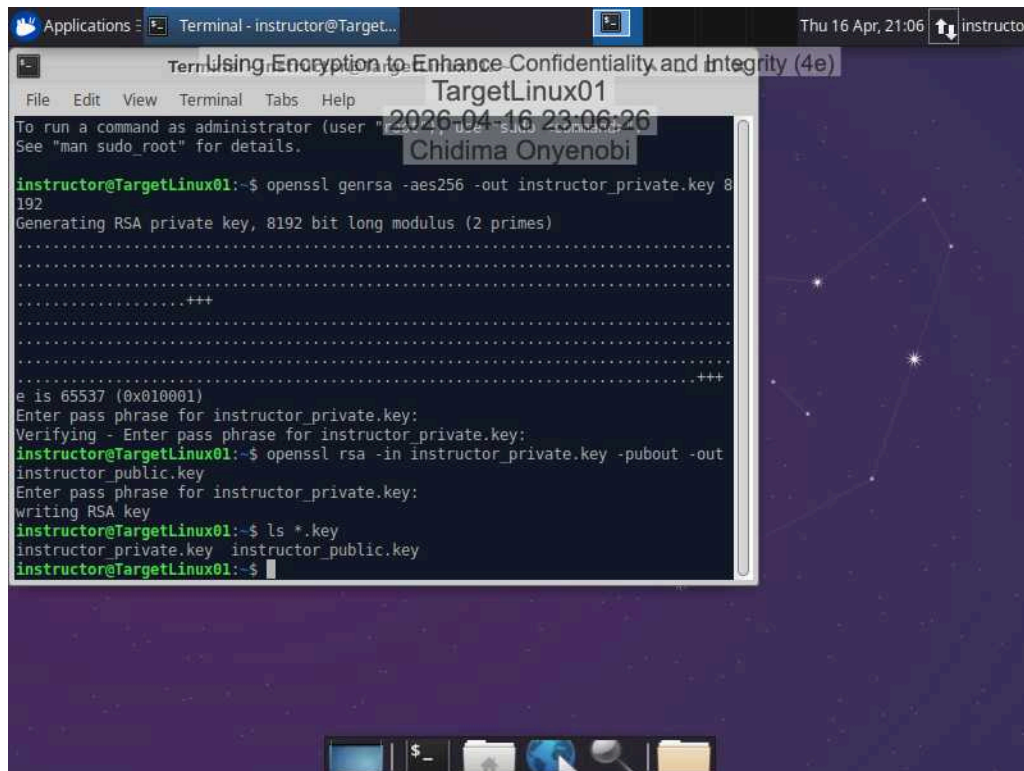
18. Make a screen capture showing the **decrypted secret-message.txt** file in Notepad.



Section 2: Applied Learning

Part 1: Create an Asymmetric Key Pair

10. Make a screen capture showing the instructor's key pair files.



The screenshot shows a terminal window titled 'Terminal - instructor@TargetLinux01'. The user is 'instructor' and the host is 'TargetLinux01'. The terminal output shows the following commands and results:

```
instructor@TargetLinux01:~$ openssl genrsa -aes256 -out instructor_private.key 8192
Generating RSA private key, 8192 bit long modulus (2 primes)
.....+++
.....+++
e is 65537 (0x010001)
Enter pass phrase for instructor_private.key:
Verifying - Enter pass phrase for instructor_private.key:
instructor@TargetLinux01:~$ openssl rsa -in instructor_private.key -pubout -out instructor_public.key
Enter pass phrase for instructor_private.key:
writing RSA key
instructor@TargetLinux01:~$ ls *.key
instructor_private.key  instructor_public.key
instructor@TargetLinux01:~$
```

A watermark '2026-04-16 23:06:26 Chidima Onyenobi' is visible over the terminal output.

Part 2: Encrypt a File Using Symmetric Encryption

11. Document the password you used to symmetrically encrypt the file.

Obj8922

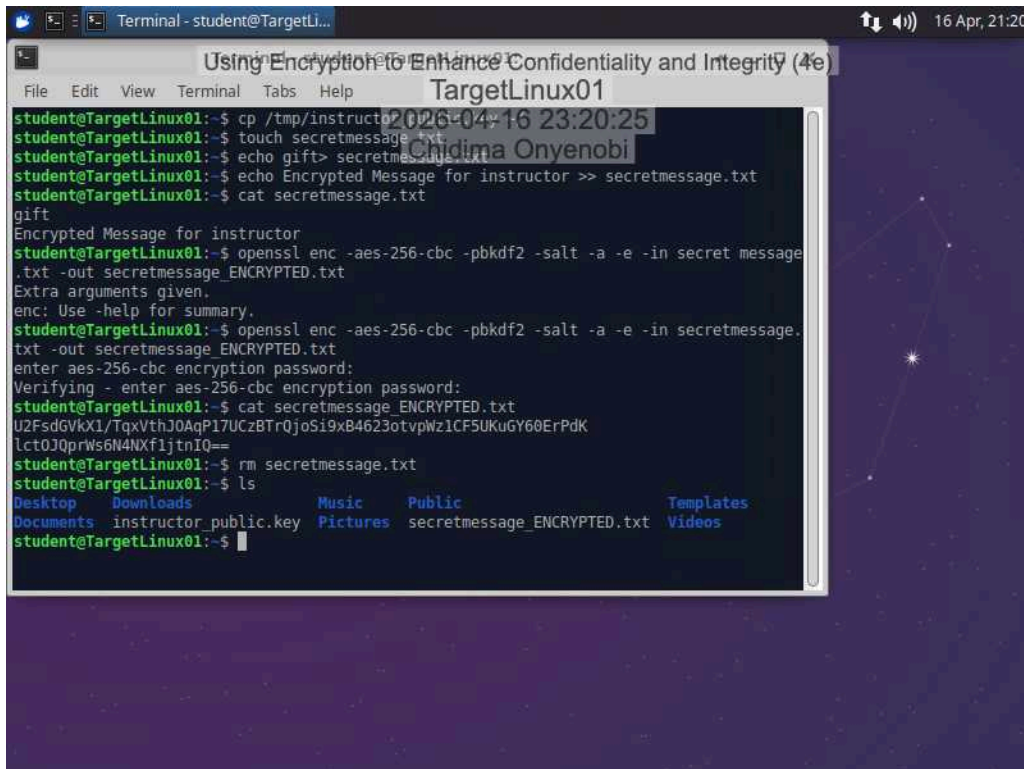
13. Make a screen capture showing the ciphertext in the `secretmessage_ENCRYPTED.txt` file.

```
student@TargetLinux01:~$ cp /tmp/instructor01/secretmessage.txt .
student@TargetLinux01:~$ touch secretmessage.txt
student@TargetLinux01:~$ echo gift > secretmessage.txt
student@TargetLinux01:~$ echo Encrypted Message for instructor >> secretmessage.txt
student@TargetLinux01:~$ cat secretmessage.txt
gift
Encrypted Message for instructor
student@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -e -in secretmessage
.txt -out secretmessage_ENCRYPTED.txt
Extra arguments given.
enc: Use -help for summary.
student@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -e -in secretmessage.
txt -out secretmessage_ENCRYPTED.txt
enter aes-256-cbc encryption password:
Verifying - enter aes-256-cbc encryption password:
student@TargetLinux01:~$ cat secretmessage_ENCRYPTED.txt
U2FsdGVkX1/TqxVthJ0AqP17UCzBTrQjoSi9xB4623otvpWz1CF5UKuGY60ErPdK
lct0JQprWs6N4NXf1jtnIQ==
student@TargetLinux01:~$
```

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16. Make a screen capture showing the output of the ls command.



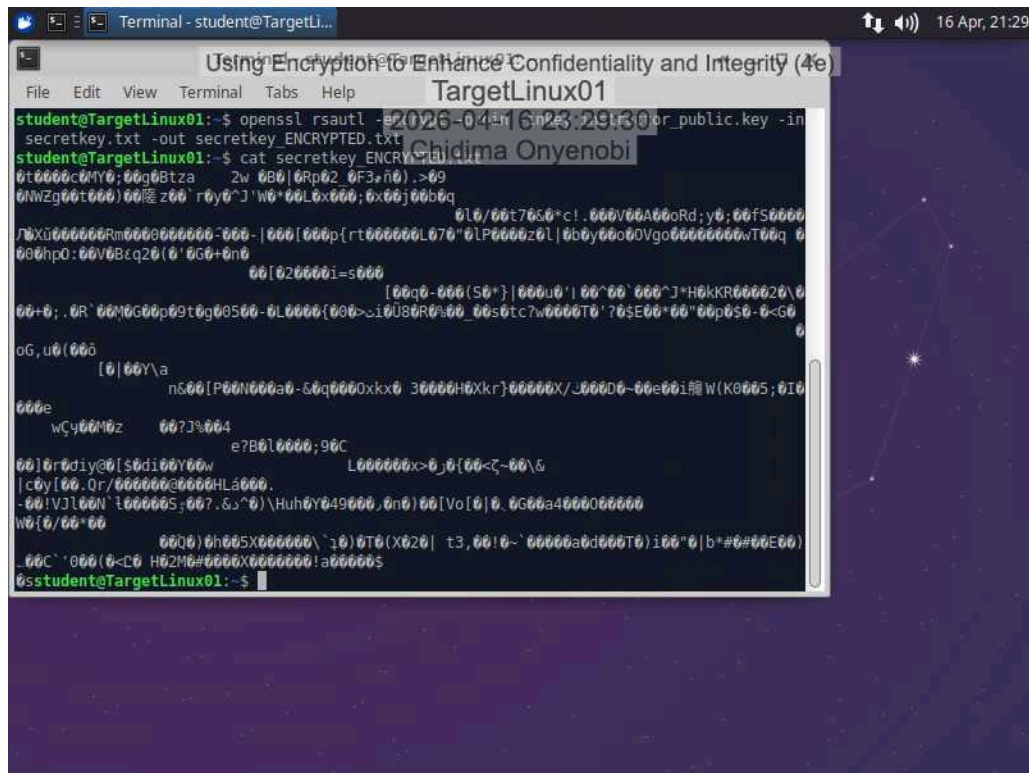
```
student@TargetLinux01:~$ cp /tmp/instructor_public.key .
student@TargetLinux01:~$ touch secretmessage.txt
student@TargetLinux01:~$ echo gift > secretmessage.txt
student@TargetLinux01:~$ echo Encrypted Message for instructor >> secretmessage.txt
student@TargetLinux01:~$ cat secretmessage.txt
gift
Encrypted Message for instructor
student@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -e -in secretmessage
.txt -out secretmessage_ENCRYPTED.txt
Extra arguments given.
enc: Use -help for summary.
student@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -e -in secretmessage.
txt -out secretmessage_ENCRYPTED.txt
enter aes-256-cbc encryption password:
Verifying - enter aes-256-cbc encryption password:
student@TargetLinux01:~$ cat secretmessage_ENCRYPTED.txt
U2FsdGVkX1/TqxVthJ0AqP17UCzBTrQjoSi9xB4623otvpWz1CF5UKuGY60ErPdK
lct0JQprWs6N4NXf1jtnIQ==
student@TargetLinux01:~$ rm secretmessage.txt
student@TargetLinux01:~$ ls
Desktop  Downloads  Music      Public      Templates
Documents instructor_public.key Pictures    secretmessage_ENCRYPTED.txt Videos
student@TargetLinux01:~$
```

Part 3: Transfer and Decrypt a File Using Hybrid Cryptography

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6. Make a screen capture showing the encrypted contents of the `secretkey_ENCRYPTED.txt` file.



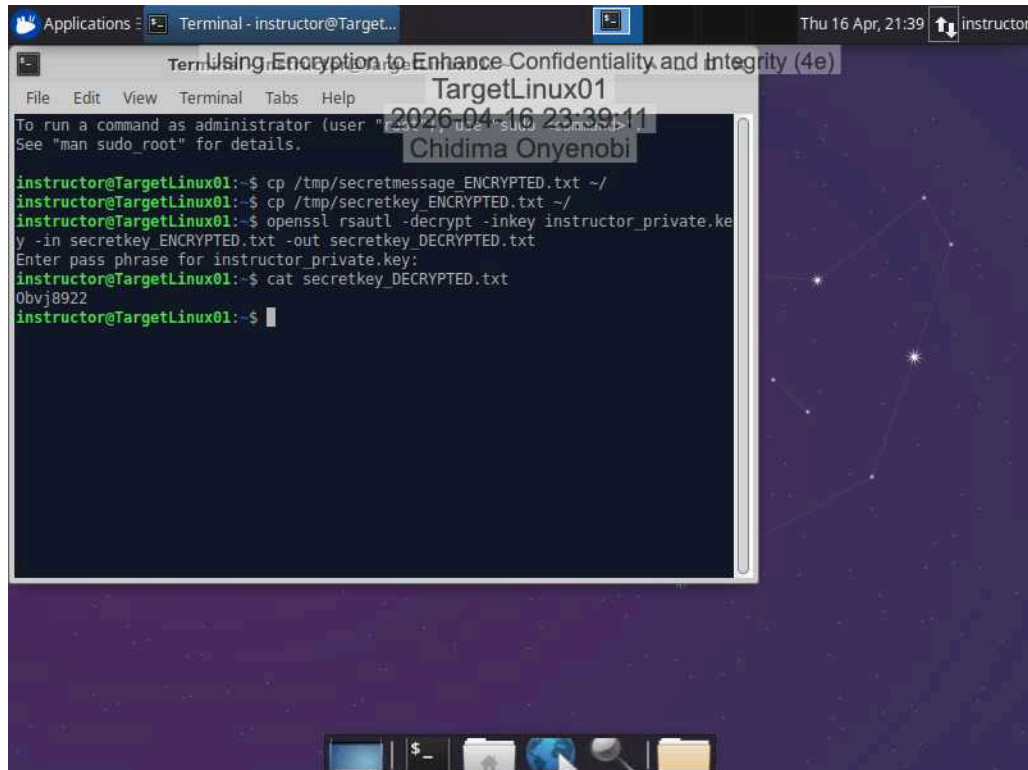
The screenshot shows a terminal window titled "Terminal - student@TargetLinux01" with a system clock of 16 Apr, 21:29. The terminal displays the following commands and output:

```
student@TargetLinux01:~$ openssl rsautl -encrypt -in secretkey.txt -out secretkey_ENCRYPTED.txt
student@TargetLinux01:~$ cat secretkey_ENCRYPTED.txt
0NwZg00t000)00z00"r0y0^J'W0*00L0x000;0x00j00b0q
0l0/00t7050*c!..000V00A00oRd;y0;00fS0000
j0XU000000Rm000000000-000-|000[000p{rt000000L070"0lP0000z0l|0b0y00o00Vgo0000000wT00q 0
000hp0:00V0B0q20(0'0G0+0n0
00[020000i=s000
[00q0-000(S0*)|000u0'|00^00'000^J*H0kKR000020\0
00+0;.0R`00\0G00p09t0g00S00-0L0000{000>0i0U0R00%00_00s0tc?w0000T0'70$E00*00"00p0$0-0<G0
0
oG,u0(000
[0|00Y\0a
n500[P00N000a0-50q0000xkx0 30000H0Xkr}00000X/0000D0-00e00i00W(K0005;0I0
000e
w0y000M0z: 007J%004
e?B0l0000;90C
00]0r0diy@0[$0di00Y00w L000000x>0_0{00<0-00\&
|c0y[00.0r/000000@0000HL0000.
-00!VJl00N`i00000S;007.&^0)\Huh0Y049000.0n0)00[Vo[0|0.0G00a4000000000
W0(0/00*00
00Q0)0h00Sx000000\`i0)0T0(X020| t3,00!0~`00000a0d000T0)i00"0|b*#0#00E00)
_00C`'000(0<00 H02M0#0000X0000000!a00000S
0sstudent@TargetLinux01:~$
```

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17. Make a screen capture showing the **decrypted contents of the secretkey_DECRYPTED.txt file.**



The screenshot shows a terminal window titled "Terminal - instructor@Target..." on a system named "TargetLinux01". The terminal displays the following commands and output:

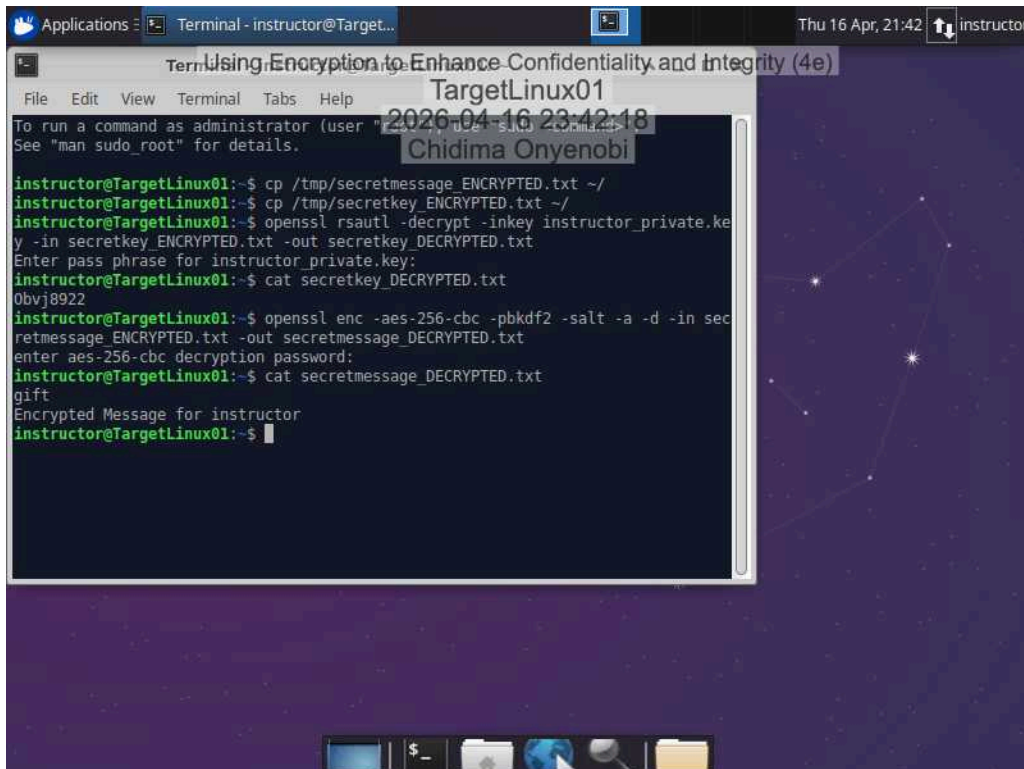
```
instructor@TargetLinux01:~$ cp /tmp/secretmessage_ENCRYPTED.txt ~/
instructor@TargetLinux01:~$ cp /tmp/secretkey_ENCRYPTED.txt ~/
instructor@TargetLinux01:~$ openssl rsautl -decrypt -inkey instructor_private.ke
y -in secretkey_ENCRYPTED.txt -out secretkey_DECRYPTED.txt
Enter pass phrase for instructor_private.key:
instructor@TargetLinux01:~$ cat secretkey_DECRYPTED.txt
0bvj8922
instructor@TargetLinux01:~$
```

Overlaid on the terminal window is a semi-transparent box containing the text: "Using Encryption to Enhance Confidentiality and Integrity (4e)", "TargetLinux01", "2026-04-16 23:39:11", and "Chidima Onyenobi". The desktop background is a dark purple space-themed wallpaper with a constellation. The taskbar at the bottom shows icons for a terminal, file manager, and other applications.

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21. Make a screen capture showing the contents of the `secretmessage_DECRYPTED` file.



The screenshot shows a terminal window titled "Terminal - instructor@Target..." with a menu bar (File, Edit, View, Terminal, Tabs, Help). The terminal output shows the following commands and results:

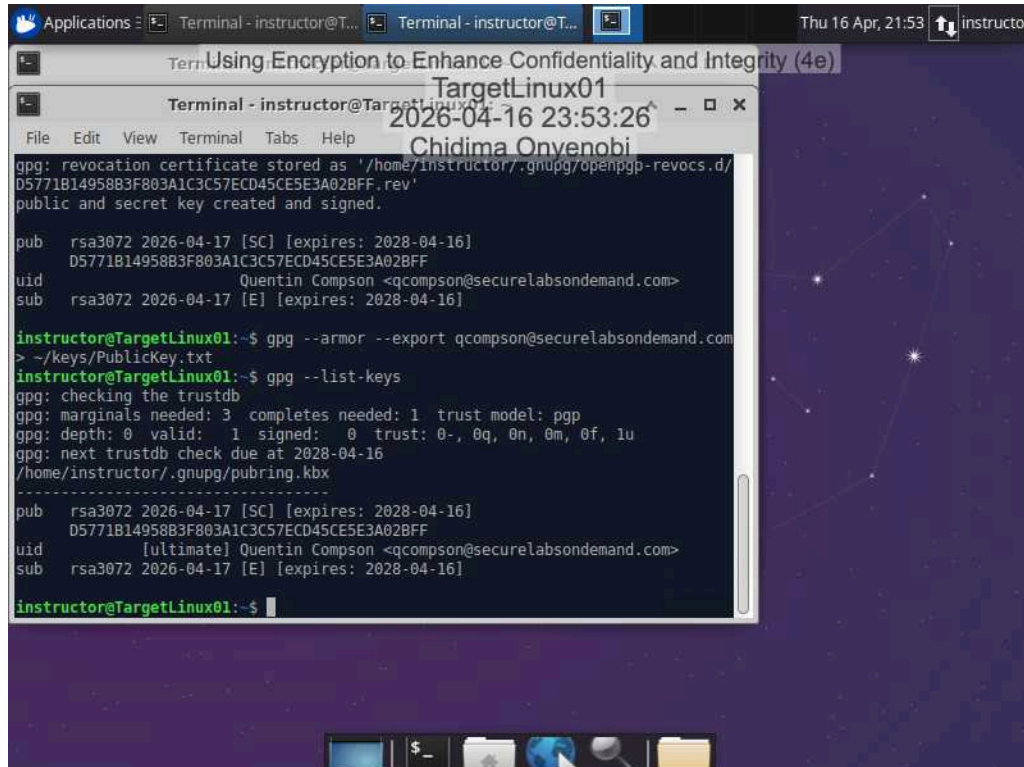
```
instructor@TargetLinux01:~$ cp /tmp/secretmessage_ENCRYPTED.txt ~/
instructor@TargetLinux01:~$ cp /tmp/secretkey_ENCRYPTED.txt ~/
instructor@TargetLinux01:~$ openssl rsautl -decrypt -inkey instructor_private.ke
y -in secretkey_ENCRYPTED.txt -out secretkey_DECRYPTED.txt
Enter pass phrase for instructor_private.key:
instructor@TargetLinux01:~$ cat secretkey_DECRYPTED.txt
0bvj8922
instructor@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -d -in sec
retmessage_ENCRYPTED.txt -out secretmessage_DECRYPTED.txt
enter aes-256-cbc decryption password:
instructor@TargetLinux01:~$ cat secretmessage_DECRYPTED.txt
gift
Encrypted Message for instructor
instructor@TargetLinux01:~$
```

Overlaid on the terminal window is a semi-transparent box containing the text: "Using Encryption to Enhance Confidentiality and Integrity (4e)", "TargetLinux01", "2026-04-16 23:42:18", and "Chidima Onyenobi". The desktop background is a dark purple space-themed image with a constellation. The system tray at the bottom shows icons for a terminal, file manager, and other applications.

Section 3: Challenge and Analysis

Part 1: Digitally Sign a Document Using GPG

Make a screen capture showing the **key fingerprint** for the key pair you generated in this part of the lab.



The screenshot shows a terminal window titled "Terminal - instructor@T..." with a dark background. The terminal output displays the following commands and results:

```
gpg: revocation certificate stored as '/home/instructor/.gnupg/openpgp-revocs.d/
D5771B14958B3F803A1C3C57ECD45CE5E3A02BFF.rev'
public and secret key created and signed.

pub  rsa3072 2026-04-17 [SC] [expires: 2028-04-16]
    D5771B14958B3F803A1C3C57ECD45CE5E3A02BFF
uid          Quentin Compson <qcompson@securelabsondemand.com>
sub  rsa3072 2026-04-17 [E] [expires: 2028-04-16]

instructor@TargetLinux01:~$ gpg --armor --export qcompson@securelabsondemand.com
> ~/keys/PublicKey.txt
instructor@TargetLinux01:~$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-04-16
/home/instructor/.gnupg/pubring.kbx
-----
pub  rsa3072 2026-04-17 [SC] [expires: 2028-04-16]
    D5771B14958B3F803A1C3C57ECD45CE5E3A02BFF
uid          [ultimate] Quentin Compson <qcompson@securelabsondemand.com>
sub  rsa3072 2026-04-17 [E] [expires: 2028-04-16]

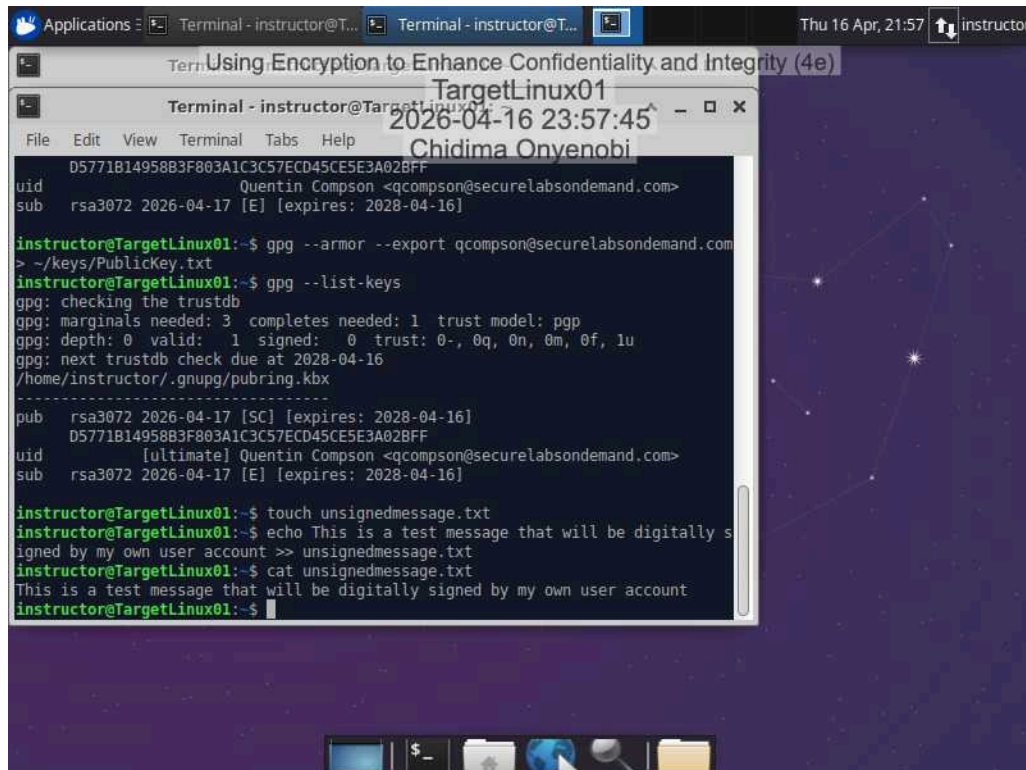
instructor@TargetLinux01:~$
```

Overlaid on the terminal window is a semi-transparent box containing the text: "Using Encryption to Enhance Confidentiality and Integrity (4e)", "TargetLinux01", "2026-04-16 23:53:26", and "Chidima Onyenobi". The desktop background is a dark purple space-themed image with a constellation of stars. The system tray at the bottom shows icons for a terminal, file manager, and other applications.

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Make a screen capture showing the contents of the unsignedmessage.txt file.

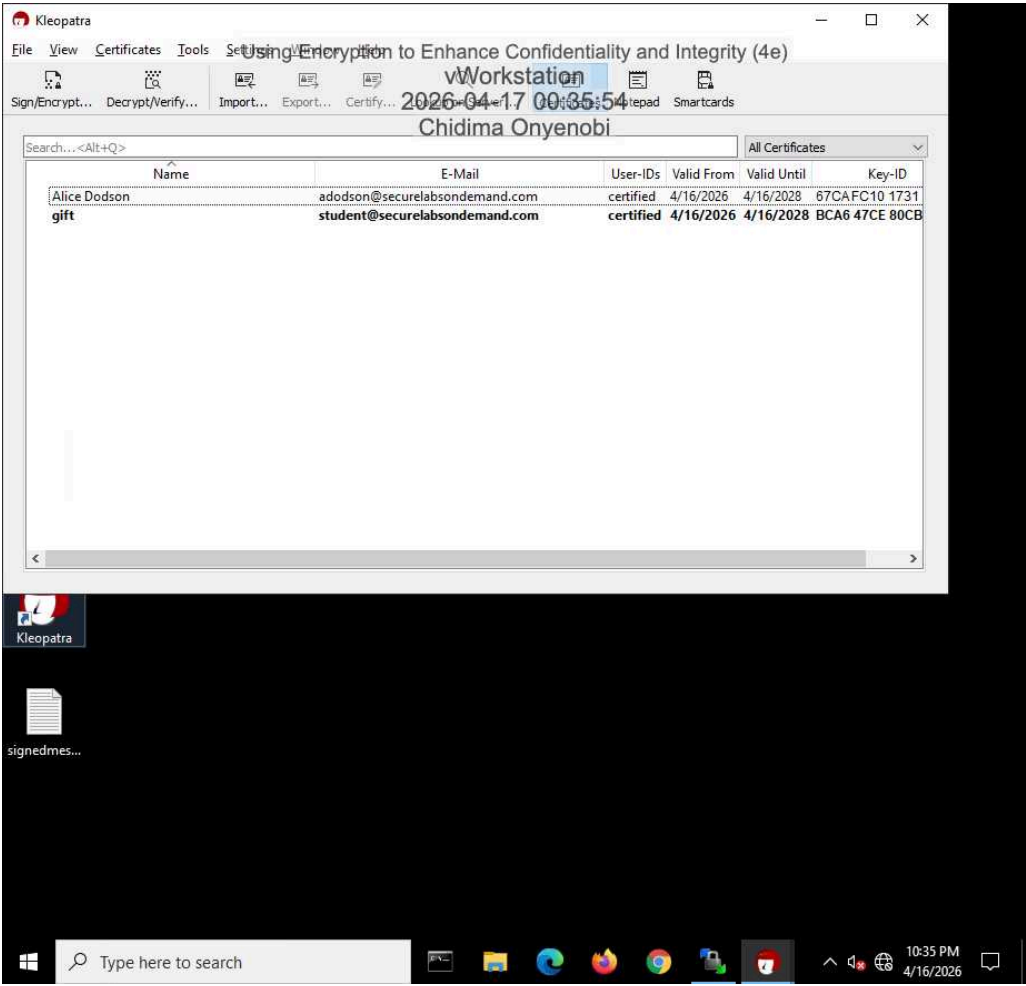


```
instructor@TargetLinux01:~$ gpg --armor --export qcompson@securelabsondemand.com
> ~/keys/PublicKey.txt
instructor@TargetLinux01:~$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-04-16
/home/instructor/.gnupg/pubring.kbx
-----
pub  rsa3072 2026-04-17 [SC] [expires: 2028-04-16]
    05771B14958B3F803A1C3C57ECD45CE5E3A02BFF
uid      [ultimate] Quentin Compson <qcompson@securelabsondemand.com>
sub  rsa3072 2026-04-17 [E] [expires: 2028-04-16]

instructor@TargetLinux01:~$ touch unsignedmessage.txt
instructor@TargetLinux01:~$ echo This is a test message that will be digitally s
igned by my own user account >> unsignedmessage.txt
instructor@TargetLinux01:~$ cat unsignedmessage.txt
This is a test message that will be digitally signed by my own user account
instructor@TargetLinux01:~$
```

Part 2: Verify the Digital Signature Using Kleopatra

Make a screen capture showing the successful signature verification on the signed message file.



i was unable to import the file